

# Smart Metro Ticket Booking System

## IDEATION PHASE

|               |                                   |
|---------------|-----------------------------------|
| Date          |                                   |
| Team ID       |                                   |
| Project Name  | Smart Metro Ticket Booking System |
| Maximum Marks | 4 Marks                           |

## 2. Empathy Map

### 2.1 Persona

**Name:** Ravi Kumar

**Age:** 27

**Occupation:** Private company employee

**Goal:** Reach office on time every day using the metro without wasting time at the ticket counter.

### 2.2 Empathy Map Canvas

| Quadrant | Observations                                                                                                                                                                                                                                                                                                                              |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SAYS     | "Why is this queue so long every morning?" "I don't have exact change for the token machine." "I wish I could just book my ticket from my phone like I book a cab." "I'm going to be late for my meeting because of this line."                                                                                                           |
| THINKS   | "There must be an easier way to do this in 2026." "What if I could see the fare before I even reach the station?" "I hope the app doesn't crash right when I need it." "Will my ticket work if the internet is slow inside the station?"                                                                                                  |
| DOES     | Reaches the station 10-15 minutes early to account for queue time. Stands in line behind other commuters. Fumbles for cash/card at the counter. Manually checks fare charts on the wall. Keeps the paper ticket carefully until the destination gate. Sometimes chooses to avoid metro entirely during rush hour and takes a cab instead. |
| FEELS    | Impatient and rushed during peak hours. Anxious about missing the train. Relieved when the queue is short. Frustrated when the token machine is out of order. Would feel confident and in control if the entire process could be completed instantly on a phone.                                                                          |

**Ravi Kumar**  
Daily metro commuter

### **Says**

Why is this queue so long?  
I don't have exact change  
Wish I could book from my phone  
I'm going to be late again

### **Thinks**

There must be an easier way  
What if I saw the fare first?  
Hope the app doesn't crash  
Will this work with slow signal?

### **Does**

Arrives 10-15 min early  
Stands in the counter queue  
Checks fare chart manually  
Keeps paper ticket carefully

### **Feels**

Impatient during peak hours  
Anxious about missing train  
Relieved when queue is short  
Frustrated by broken machines

### **Pains**

Time lost standing in queues  
Risk of missing the train  
Lost or misplaced tokens

### **Gains**

Book a ticket in under a minute  
Instant, accurate fare  
QR ticket, no token needed

## **2.3 Pains**

- Losing 5-15 minutes daily standing in queues, adding up to significant time loss over a month.
- Risk of missing a train due to slow manual processing at counters.
- Physical tokens can be misplaced, dropped, or forgotten, causing re-purchase.
- No visibility into fare, route, or estimated travel time before arriving at the station.
- Poor experience for first-time riders, tourists, elderly, and differently-abled commuters who need extra assistance that counters may not always provide efficiently.

## **2.4 Gains (What Success Looks Like)**

- Ability to book a ticket in under a minute from anywhere, before reaching the station.
- Instant, accurate, automatically calculated fare with no manual errors.
- A digital QR ticket that can be scanned directly at the gate — no physical token needed.
- Automatic notification/confirmation of the booking, giving peace of mind.
- A consistent, accessible experience for every type of commuter, including elderly and differently-abled passengers.
- Reduced crowding at stations, making the overall commute smoother for everyone.

## 2.5 Insight Summary

The empathy map shows that the core commuter frustration is **time lost to manual, in-person processes** and **uncertainty about fare and queue length**. Commuters clearly want a fast, self-service, mobile-friendly way to book and pay for tickets before they even arrive at the station. This directly supports building a ServiceNow Service Catalog-based booking flow that automates fare calculation and delivers an instant QR-code ticket, removing the counter dependency altogether.